



XVII Ibero-American Interuniversity Mathematics Competition - Turrialba, Costa Rica

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– Day 1

- 1** Consider two sets $P, Q \subseteq \mathbb{R}^2$ that satisfy the following three conditions:
1. $(n, 1) \in P$ for every $n \in \mathbb{Z}$.
 2. If $(a, b) \in P$, then $(a, 0) \in Q$.
 3. If a trapezoid has two sides parallel to the y -axis, with two of its vertices in P and the other two vertices in Q , then the point of intersection of its diagonals is also in P .
- Prove that $\mathbb{Q} \times \{0\} \subseteq Q$.

- 2** Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$F(x, y) = (x + y - x^3, y - y^3).$$

Define $F^1 = F$, and

$$F^{n+1} = F \circ F^n \quad \text{for } n \geq 1.$$

- a) Prove that if $|x| \leq \frac{1}{2}$ and $|y| \leq \frac{1}{2}$, then

$$\lim_{n \rightarrow \infty} F^n(x, y) = (0, 0)$$

- b) Prove that for any $K > 0$ and $\varepsilon > 0$, there exist a positive integer n and a point $(x_0, y_0) \in \mathbb{R}^2$ with $0 < |(x_0, y_0)| < \varepsilon$ such that

$$|F^n(x_0, y_0)| > K \cdot |(x_0, y_0)|$$

- 3** Let E be an ellipse with semi-axes $a > b$ and let B be an endpoint of the minor axis of the ellipse. For every point P on the ellipse, define $m(P)$ as the greatest distance from P to any other point on the ellipse; that is, define the function

$$m(P) = \max_{Q \in E} |PQ|,$$

where $|PQ|$ denotes the distance from P to Q .

- Prove that $m(B) \leq m(P)$ for every point P on the ellipse.
- Determine the value of $m(B)$ as a function of a and b .

– Day 2

- 4 Find infinitely many differentiable functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $f(20, 25) = 2025$ and, for all $x, y \in \mathbb{R}$, the following identity holds:

$$(f(x, y))^3 + (f_x(x, y))^3 + (f_y(x, y))^3 = 3f(x, y)f_x(x, y)f_y(x, y),$$

where f_x and f_y denote the partial derivatives of f with respect to x and y , respectively.

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- 5 Let n be a positive integer. In the space of $n \times n$ matrices with real entries, consider the subset M_n of matrices whose entries belong to the set $\{-1, 1\}$.
Prove that at least 25

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- 6 Prove that there exist positive real numbers C and t , with $t > 1$, such that if $n \geq 2$ and S is a set of $2n$ points in the plane in general position, with n red points and n blue points, then there are at least Cn^t triangles with two red vertices and one blue vertex whose interior contains no point of S .

Note: A set of points is said to be in general position if no three points are collinear.
